Automotive Climate Control

Tsutomu Tabe, Katsumasa Matsui, Tohru Kakehi, and Masahiro Ohba

ABSTRACT: This paper deals with automotive climate control using modern control theory. Automobile compartment temperature is controlled by varying compressor capacity and the position of the air mixing damper. The mathematical model of the system is obtained using a statistical identification method, and this system model is reduced to lower order using minimal realization. The experimental results show robust control performance under climate changes causing parameter variations or disturbances.

Introduction

Since an automobile is operated in various weather conditions, such as scorching heat and downpours, compartment temperature is constantly affected by environmental changes. An air-conditioning system must maintain a comfortable inside temperature despite these changes. However, an air-conditioning system inevitably uses energy, which increases car fuel consumption. Because fuel consumption is a concern, this energy must be minimized. Therefore, an entirely new control procedure is needed to resolve the contradictions of low energy consumption and a pleasant driving climate.

The additional complexity needed for more sophisticated control can be implemented using modern-day automotive electronics such as microprocessors, which are beginning to be used in automotive air-conditioning systems. However, there are inefficiencies inherent in the traditional compressors. The present authors adopt a recently developed variable-capacity compressor into an air-conditioning system for climate control in order to resolve the conflicting demands for driving comfort and low energy consumption. Computer-aided design (CAD) has simplified control systems design, paving the way for implementation of modern control theory [1]. The present study applies modern control theory to design automotive climate control, and presents representative experimental results to evaluate performance.

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Tsutomu Tabe, Katsumasa Matsui, Tohru Kakehi, and Masahiro Ohba are with the Research and Development Department (Electronics), Nippondenso Co., Ltd., 1-1 Showa-cho, Kariya-city, Aichi 448, Japan.

Air-Conditioning System

Figure 1 shows an overview of the air-conditioning system under study. The outside air drawn into the air conditioner is first cooled by heat exchange within the evaporator. The cooling capability of the evaporator is controlled by changing the capacity of the variable compressor. Since the compressor is powered by the engine, the lower the compressor capacity, the lower the load on the engine, and the lower the energy consumption.

The outside air cooled inside the evaporator flows in two directions because of the separation effected by the air mixing damper. One part flows as cool air, while the other is reheated within the heater core by heat exchange. Thus, the air mixing damper adjusts the amount of outside air to be reheated. The cool and warm air are remixed and blown into the car compartment through outlets. Hence, the temperature inside the automobile is determined by the amount of cooling by the compressor, the amount of reheating from the air mixing damper, and the heat balance inside and outside the compartment.

Mathematical Model

A mathematical model of an air-conditioning system can be obtained from physics by starting with a heat equation. However, a car undergoes various climate changes, such as outside air temperature and amount of solar radiation, so operating conditions are constantly changing. Therefore, the physically obtained model of the air-conditioning system will be quite complex and parameters must be tested and properly confirmed.

Thus, the present authors assume that changes of weather and driving conditions are disturbance variables, and the authors linearized the system. Figure 2 presents the simplified system with compressor capacity and air mixing damper position as input variables, compartment temperature as the controlled variable, and sensor output as the output variable.

The order of this simplified model and the parameters are determined by a statistical identification method. In the method, the dynamics are assumed to be a linear system, and are identified by test signals and measured output signals under some climate and driving conditions. For the test signals, an irregular signal such as the air mixing damper position indicated in Fig. 3 is employed. Figure 4 shows the output signal (measured temperature at the upper outlet) and simulation result. The system identification re-

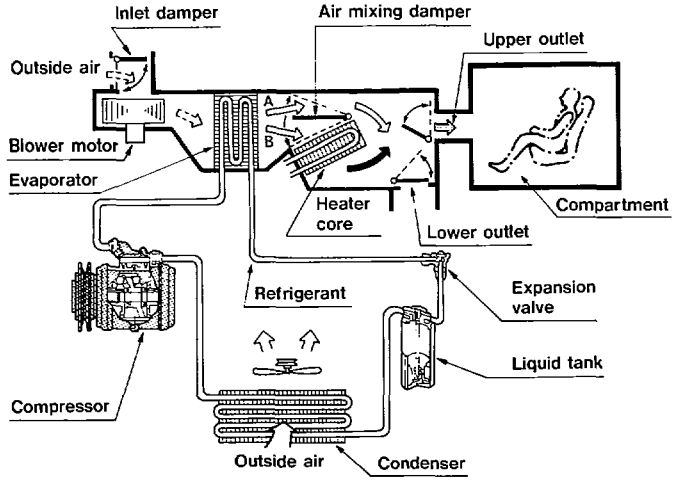


Fig. 1. Overview of air-conditioning system.

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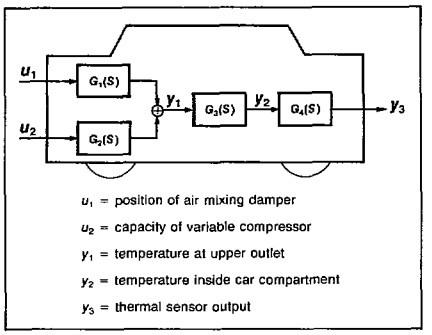


Fig. 2. Block diagram of air-conditioning system.

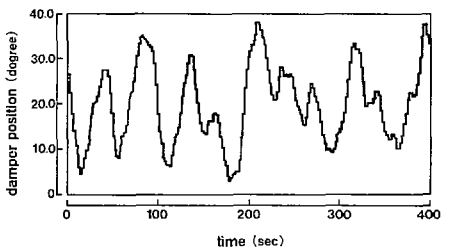


Fig. 3. Test signals for modeling of dynamics.

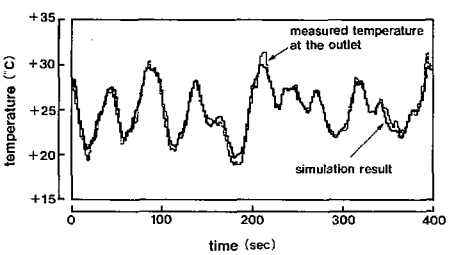


Fig. 4. Result of system identification.

sults are found to correspond to actual performance. In this way, the dynamics are obtained, and the air-conditioning system is defined as a ninth-order model. Table 1 gives the parameters for the state equation and the output equation of the ninth-order system. There are two control variables and three output variables as well as the nine state variables.

Model Reduction

Digital control using modern control theory requires much numerical calculation. To decrease calculations, the reduction of the model order is carefully considered. Many different methods have been proposed for model reduction. The authors employ the numerically stable algorithm for practically minimal realization of Kaneda and Matsui [2] for the reduction of the model.

From this analysis, it is concluded that the ninth-order air-conditioning system model

Table 1
Parameters for State Equation and Output Equation with Ninth-Order Model—with Two Control Variables and Three Output Variables

$$\dot{x} = Ax + Bu,\ y = Cx$$

$$A = \begin{bmatrix} 0 & -0.05 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1.51 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.03 & -1.04 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.004 & 0 & -0.03 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1.04 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -0.03 & 0 & 0 & 0 \\ 0 & 0 & 0.03 & 0.03 & 0.03 & 0.03 & -0.28 & 0 & 0 \\ 0 & 0 & 0.005 & 0.005 & 0.005 & 0.005 & 0 & -0.02 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.06 & 0.06 & -0.06 \end{bmatrix}$$

$$B^T = \begin{bmatrix} -0.22 & -1.46 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 14.4 & 0.14 & 0 & 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

can be approximated by a third- or fourth-order system. Figure 5 gives the temperature step response of the nth-order model (n = 2, 3, 9), and the response of the third-order model is virtually the same as that of the original ninth-order model response. Thus, the third-order model is adopted as the design model. Table 2 gives the parameters for the state equation and the output equation of this model. With the reduced-order model, there are two control variables and one output variable as well as three state variables.

Design of the Control System

Compliance of the vehicle interior temperature with a target temperature is required despite external disturbances and model uncertainty. Thus, a linear control system is devised that consists of a control law, an observer, and an integrator. CAD [1] is used in the control system design. The following equations specify the servo system with the integral amount $z(t)$ of the output deviation $e(t)$ as the fourth state variable, and the matrices A, B, and C defined in Table 2.

$$\begin{bmatrix} \dot{x}(t) \\ \dot{z}(t) \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \begin{bmatrix} x(t) \\ z(t) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(t) + \begin{bmatrix} 0 \\ -1 \end{bmatrix} y^*(t) \quad (1)$$

$$y(t) = [C \ \ 0] \begin{bmatrix} x(t) \\ z(t) \end{bmatrix} \quad (2)$$

$$e(t) = y(t) - y^*(t) \quad (3)$$

$$x(t)^T = [x_1(t)\ x_2(t)\ x_3(t)]$$

The control input variables are the air mixing damper position $u_1(t)$ and the compressor capacity $u_2(t)$. The scalar output $y(t)$ is the sensor output of the compartment temperature, and $y^*(t)$ is the temperature command. The control law is the feedback of a linear combination of all the states, that is,

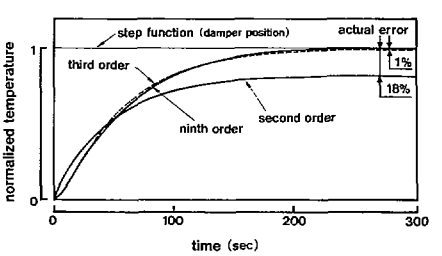


Fig. 5. Comparison of temperature response in nth-order model.

Table 2
Parameters for State Equation and Output Equation with Reduced Third-Order Model—with Two Control Variables and One Output Variable

$$\dot{x} = Ax + Bu,\ y = Cx$$

$$A = \begin{bmatrix} -0.02 & 0.008 & 0.001 \\ 0.04 & -0.12 & -0.05 \\ -0.006 & 0.04 & -0.02 \end{bmatrix} \quad B = \begin{bmatrix} -0.0003 & 0.013 \\ -0.00003 & -0.013 \\ 0.0015 & 0.0018 \end{bmatrix}$$

$$C = [0.008 \quad 0.007 \quad 0.002]$$

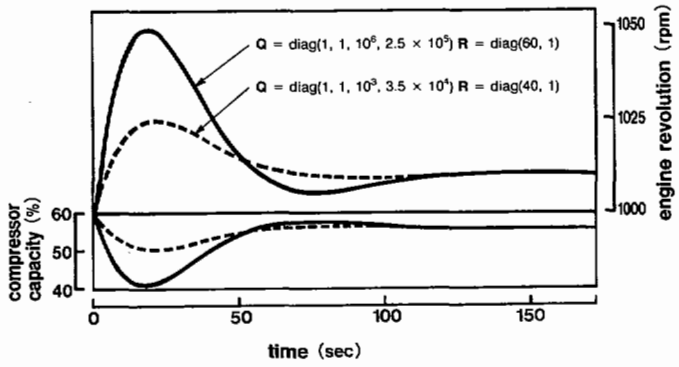


Fig. 6. Effect of compressor capacity on engine revolution.

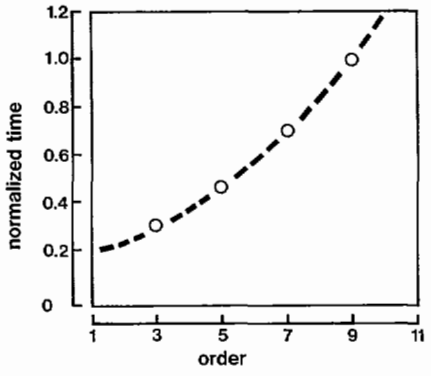


Fig. 8. Calculation time for each order model.

$$u(t) = -F_1 x(t) - F_2 z(t) \quad (4)$$

We solve the control problem by choosing $u(t)$, which minimizes the cost function J.

$$J = \int_0^\infty (\|\delta x(t)\|_Q^2 + \|\dot{u}(t)\|_R^2)\, dt \quad (5)$$

$$\delta x(t) = x(t) - x(\infty) \quad (6)$$

By the use of optimal control theory, the control gains F_1 and F_2 in Eq. (4) are expressed as

$$[F_1\ F_2] = R^{-1}B^T P \quad (7)$$

Here P is the solution of the following Riccati equation, and Q and R are the weighting matrices in Eq. (5).

$$A^T P + PA + Q - PBR^{-1}B^T P = 0 \quad (8)$$

In solving the control problem, two performance criteria are used.

1. In relation to step variations in the temperature command, the settling time for the compartment temperature must be within 60 sec without overshoot.
2. The overshoot of the compressor capacity is kept within 100 percent to prevent any deterioration in drivability, because changes in the compressor capacity affect engine revolution as can be seen in Fig. 6.

As a result, the weighting matrices Q and R are selected as

$$Q = \text{diag}\,[1 \quad 1 \quad 10^3 \quad 3.5 \times 10^4]$$

$$R = \text{diag}\,[40 \quad 1] \quad (9)$$

The control gain shown in Table 3 is thereby obtained from Eqs. (7) and (8). Since not all states are measured, we must design an observer. One output $y_3(t)$ can be measured, thus, using the reduced-order observer of Luenberger, $x(t)$ is estimated. The observer poles are selected so as to be twice as fast as the control poles by simulation, based on noise rejection characteristics.

Table 3
Control Gain Matrices

$$F_1 = \begin{bmatrix} -10.74 & -2.92 & 0.82 \\ 6.35 & 1.75 & -0.48 \end{bmatrix} \qquad F_2 = \begin{bmatrix} -47.36 \\ 28.62 \end{bmatrix}$$

Figure 7 shows a discrete-time control system. The integrator is constructed from the accumulative calculation presented in Eq. (10), where T is the sampling period.

$$z(k) = z(k-1) + Te(k) \quad (10)$$

Experimental Results

The servo system under study is implemented with an 8-bit microprocessor. The servo controller with the aid of the temperature sensor determines the drive signals of the variable compressor and air mixing damper, so that the car compartment temperature will attain the desired temperature commanded by the driver. The drive signals are obtained by floating-point calculations using the control law, the reduced observer, and the integration. The calculation time for the third-order model, as can be seen in Fig. 8, is about 30 percent of that for the ninth-order model. Thus, a lower grade microprocessor can serve as the controller.

The performance of the air-conditioning system is investigated by testing with the servo controller. Figure 9 indicates the step responses of the car compartment temperature. Weather conditions are the same as at the time of the system identification. The compartment temperature attains the target temperature within 60 sec without any overshoot. The compressor capacity is varied 5 percent in relation to every two-degree change in the temperature command. As a result, the control is achieved with no wasted energy. However, it is beyond the scope of the present study to determine whether the capacity variation is ideal from an energy-consumption standpoint.

Next, the robustness of the control is examined when weather conditions change. Figure 10 indicates the compartment temperature step responses from 26°C to 24°C, under (a) the same conditions at the time of the system identification, (b) when the outside temperature changes, and (c) when solar radiation conditions are added. Robustness is indicated, with only about a 20 percent difference in response time.

Next, the control performance is investi-

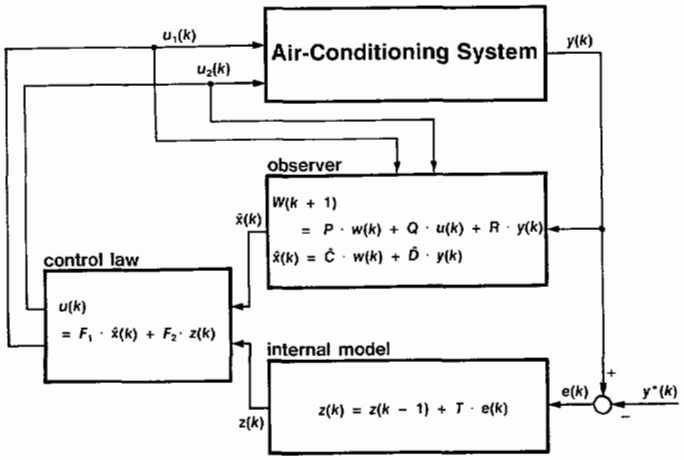


Fig. 7. Block diagram of automotive climate control system.

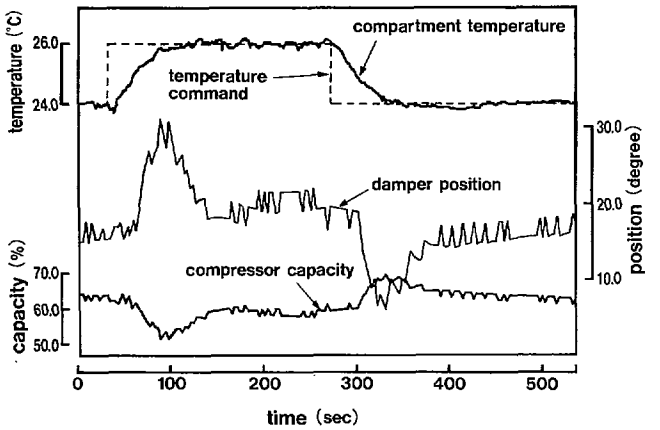


Fig. 9. Step response of system and variation of input variables.

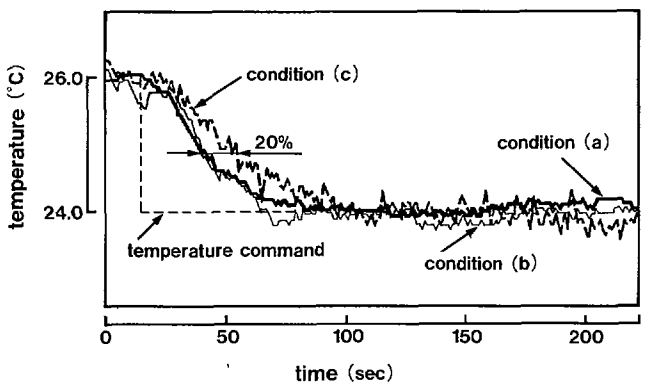


Fig. 10. Comparison of step response of system under different climate conditions. Experimental conditions: (a) outside temperature +30°C; (b) outside temperature +20°C; (c) outside temperature +30°C, solar radiation 1000 kcal/m^{2}h.

gated when solar radiation disturbance takes place in the compartment, as shown in Fig. 11. The compartment temperature is controlled so as to remain within 26°C ± 0.3°C.

The compressor capacity changes in proportion to the amount of solar radiation. The new system enables stable control even without a solar radiation sensor.

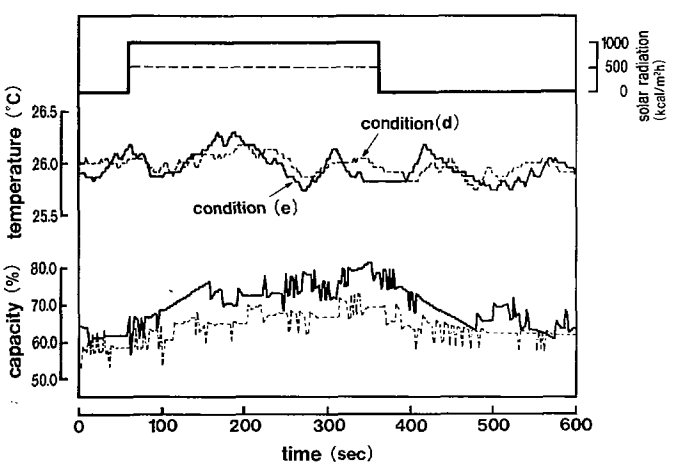


Fig. 11. System performance under solar radiation disturbance. Experimental conditions: (d) solar radiation, 500 kcal/m^{2}h; (e) solar radiation, 1000 kcal/m^{2}h.

Conclusions

Automotive climate control was performed using modern control theory in an air-conditioning system with a variable-capacity compressor. As a result, even without detecting changes in weather conditions, a robust climate control was achieved under changes causing parameter variables or disturbances. Also, by means of the variable compressor capacity, climate control was possible without compromising driving comfort or energy consumption.

The air-conditioning system model was obtained using a statistical identification method to determine model order and parameters. The authors used Kaneda's and Matsui's minimal realization method to reduce this model. It was demonstrated that a variable compressor can be implemented using modern control. However, a quantitative analysis from a low energy-consumption standpoint has yet to be undertaken. Therefore, further study is required to achieve the goal of low energy consumption.

Acknowledgment

Sincere gratitude is expressed to Professor M. Kaneda (Takuma Radio Technical College) for his incisive suggestions and assistance in the course of this study.

References

[1] M. Kaneda and H. Kuraoka, "A CAD System on a Personal Computer for Designing an Actual Control System," *Proc. IECON*, vol. 1, pp. 237–242, 1984.

[2] M. Kaneda and K. Matsui, "Numerically Stable Algorithm for Practically Minimal Realization and Its Application," *Proc. IECON*, vol. 2, pp. 593–597, Nov. 1985.



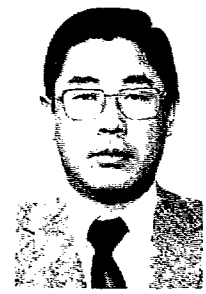
Tsutomu Tabe was born in Osaka, Japan, in April 1933. He received the B.Eng. degree in electrical communication engineering from Osaka University in 1957. Then he joined Oki Electric Industry Co., Ltd., and worked toward the development of digital computers. In 1972, he joined Nippondenso Co., Ltd., and has done research and development work in the automotive electronics field as a Manager. Presently, he is a General Manager of the Research and Development Department. His current interests include implementation of modern control theory for sophisticated automotive control systems.



Katsumasa Matsui was born in Osaka, Japan, in January 1955. He received the B.E. and M.E. degrees in electronic engineering from Osaka University in 1978 and 1980, respectively. His graduate research was concentrated in the area of multimicroprocessor systems on the data-flow scheme. He joined Nippondenso Co., Ltd., as a Research and Development Engineer in 1980. From 1980 to 1982, he designed digital control systems for automotive products, and has done research in applications of modern control theory to automotive engineering since 1983. His current interests are robot control and high-speed controllers for implementation of modern control theory.



Tohru Kakehi was born in Nagoya, Japan, in September 1958. He received the B.E. degree in telecommunication engineering from Tokyo Denki University in 1982. His graduate research was a simple compiler design of a personal computer. He joined Nippondenso Co., Ltd., as a Research and Development Engineer in 1982. From 1982 to 1984, he developed digital controllers for automotive engines. He has done research in applications of modern control theory to automotive engineering since 1984. His current interest is a microprocessor-based high-speed real-time controller for implementation of modern control theory.



Masahiro Ohba was born in Beijing, China, in September 1944. He received the B.E. degree in electronics and communication engineering from Waseda University in 1968. From 1968 to 1976, he worked as a Researcher at Victor Company of Japan, Ltd. He joined Nippondenso Co., Ltd., as a Research and Development Engineer in 1976, and has developed digital controllers for automotive systems. He is currently a Manager of the Research and Development Department (Electronics), and is working on the application of modern control theory to various automotive control systems.

Conference Calendar continued from page 14.

nica de Valencia, P.O. Box 22012, 46071-Valencia, Spain, phone: (34) (6) 360 40 41, Telex: 62808 UPV.

IFAC/IFIP/IMACS International Symposium on Theory of Robots, Dec. 3-5, 1986, Vienna, Austria. Contact: Dr. P. Kopacek, ÖPWZ, P.O. Box 131, A-1014 Vienna, Austria.

IEEE Control Systems Society 1986 Conference on Decision and Control
Dec. 10-12, 1986
Athens, Greece
General Chairman:
Prof. Anthony Ephremides
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IFAC Symposium on Automation and Instrumentation for Power Plants, Dec. 15-17, 1986, Bangalore, India. Contact: Dr. M. Ramamoorty, % IFAC86, The Institution of Engineers (India), Karnataka State Centre, Ambedkar Veedhi, Bangalore-560 001, India.

1987 Listings

1987 International Symposium on Intelligent Control
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1987 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Apr. 6-9, 1987, Registry Hotel, Dallas, Texas. Technical Program Chairman: Dr. Masud M. Arjmand, Texas Instruments, Inc., P.O. Box 226015, MS 238, Dallas, TX 75266, phone: (214) 995-0395.

1987 IEEE International Symposium on Circuits and Systems, May 4-7, 1987, Philadelphia, Pennsylvania. Technical Conference Chairman: Dr. S. D. Bedrosian, Department of Electrical Engineering, University of Pennsylvania, Philadelphia, PA 19104, phone: (215) 898-8518.

1987 American Control Conference
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IMACS/IFAC International Symposium on Modeling and Simulation of Distributed Parameter Systems, Oct. 6-9, 1987, Hiroshima, Japan. Send two copies of 300-word abstract by Dec. 1, 1986, to Dr. Tanehiro Futagami, IMACS-DPS'87-Hiroshima, Dept. of Civil Engineering, Hiroshima Institute of Technology, Itsukaichi, Saeki, Hiroshima, 731-51, Japan.